

# One Wrong Move: Pattern And Symbol Systems

**Report:** 007-pattern-symbol-systems-report.pdf

**Date:** 2026-05-02

**Branch:** pattern-symbol-systems

## Summary

Expanded One Wrong Move from a small abstract puzzle pool into an 18-type source-world pattern system while preserving the mobile-first static app, paused round flow, active-time-only timer, lower-is-better scoring, daily deterministic generation, and same-session break-signature variation.

The new direction is: symbols are rule carriers, not decoration. Cards carry rank and suit rules, chess pieces carry movement rules, Go stones carry liberty rules, logic gates carry truth tables, and dominoes/dice carry numeric constraints.

## Repo State Summary

- Working folder: `/Users/kevinhegg/Desktop/one-wrong-move`
- Git repo confirmed; current branch: `pattern-symbol-systems` .
- Nested repo check at start found only `./.git` .
- Remote: `git@github.com:KevinHegg/one-wrong-move.git` .
- Lower-is-better scoring commit was already present before this branch.

## Git Commands Run

```
pwd
git status --short
git branch --show-current
git log --oneline -5
git remote -v
```

```
find . -maxdepth 3 -name .git -type d
ls -la
node --check public/scoring.js
node --check public/puzzles.js
node --check public/game.js
node scripts/validate-scoring.js
node scripts/validate-puzzles.js
git diff --check
git switch -c pattern-symbol-systems || git switch pattern-symbol-systems
```

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## Files Created Or Changed

- `public/symbols.js` created: structured cards, chess, checkers, Go, logic, dice/domino, ecology, direction, clock, and abstract packs.
- `public/puzzles.js` rebuilt: rich puzzle model, 18 production types, daily selector, validators, break signatures.
- `public/game.js` updated: compound symbol rendering, symbol chips, source-world labels, accessible cell labels.
- `public/style.css` updated: card, chess, Go, logic, domino, dice, animal, lab, and compound token styling.
- `public/index.html` updated: loads symbols before puzzles and shows symbol chips.
- `public/lab.html` and `public/lab.js` created: development puzzle lab.
- `scripts/validate-puzzles.js` expanded and `scripts/validate-pattern-systems.js` created.
- `README.md`, `docs/game-design.md`, and `docs/pattern-systems.md` updated/created.
- `docs/codex-reports/screenshots/007-*.png` created.
- `docs/codex-reports/007-pattern-symbol-systems-report.pdf` created.

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## New Symbol System

`window.OWM_SYMBOLS` now provides structured symbol packs. Symbols include id, glyph, label, short label, category, tags, fallback, ARIA label, and useful metadata such as card rank/suit, chess movement, Go adjacency, logic values, and direction angles.

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## New Pattern Engine

Every generated round now returns a richer object with identity, source world, difficulty, cognitive skill, symbols, briefing, example, board, answer index, explanation, wrong-tap hint,

break signature, validator, and evidence. Board cells support primary glyphs, compound symbols, corner labels, class names, ARIA labels, values, and interactivity flags.

## Production Puzzle Types

| Type            | Source world   | Difficulty | Rule                             |
|-----------------|----------------|------------|----------------------------------|
| Rule Rows       | Symbol Grammar | 1          | Shifted row grammar              |
| Conveyor Shift  | Symbol Grammar | 2          | Row-to-row shift                 |
| Rotation Logic  | Compass        | 2          | Fixed direction rotation         |
| Latin Trap      | Symbol Grammar | 2          | Row/column one-of-each           |
| Pair Pact       | Symbol Grammar | 2          | Partner pairs                    |
| Path Rhythm     | Map            | 3          | Movement rhythm                  |
| Mirror Trap     | Symbol Grammar | 3          | Mirror transformation            |
| Card Straight   | Cards          | 1          | Rank progression and suit rhythm |
| Suit Cycle      | Cards          | 2          | Shifted suit cycle               |
| Knight Path     | Chess          | 3          | Legal knight moves               |
| Chess Attack    | Chess          | 4          | Piece attack vectors             |
| Go Liberties    | Go             | 4          | Orthogonal liberties             |
| Logic Gate Row  | Logic          | 2          | Truth-table outputs              |
| Domino Chain    | Dominoes       | 2          | Matching domino halves           |
| Dice Sum        | Dice           | 2          | Small arithmetic totals          |
| Checkers Jump   | Checkers       | 3          | Diagonal movement                |
| Animal Food Web | Ecology        | 2          | Food-chain order                 |
| Compass Rose    | Compass        | 2          | Bearing rotation                 |

## Candidate Pattern Systems

| Candidate      | Source world     | Status      |
|----------------|------------------|-------------|
| Rule Rows      | Abstract grammar | Implemented |
| Conveyor Shift | Abstract grammar | Implemented |
| Rotation Logic | Compass          | Implemented |

|                    |                 |             |
|--------------------|-----------------|-------------|
| Latin Trap         | Latin square    | Implemented |
| Pair Pact          | Pair grammar    | Implemented |
| Path Rhythm        | Map path        | Implemented |
| Mirror Trap        | Mirror mapping  | Implemented |
| Card Straight      | Playing cards   | Implemented |
| Suit Cycle         | Playing cards   | Implemented |
| Knight Path        | Chess           | Implemented |
| Chess Attack       | Chess           | Implemented |
| Go Liberties       | Go / baduk      | Implemented |
| Logic Gate Row     | Digital logic   | Implemented |
| Domino Chain       | Dominoes        | Implemented |
| Dice Sum           | Dice            | Implemented |
| Checkers Jump      | Checkers        | Implemented |
| Animal Food Web    | Ecology         | Implemented |
| Compass Rose       | Compass / clock | Implemented |
| Poker Hand Grid    | Playing cards   | Backlog     |
| Checkmate Net      | Chess           | Backlog     |
| Go Capture Race    | Go / baduk      | Backlog     |
| Circuit Trace      | Electronics     | Backlog     |
| Train Routes       | Transit maps    | Backlog     |
| Music Measure      | Music           | Backlog     |
| Chemistry Valence  | Chemistry       | Backlog     |
| Sports Formation   | Sports          | Backlog     |
| Calendar Week      | Calendar        | Backlog     |
| Language Pattern   | Linguistics     | Backlog     |
| Map Compass Walk   | Maps            | Backlog     |
| Set Attribute Grid | Set-style logic | Backlog     |

## Implemented New Types

Implemented production-ready Card Straight, Suit Cycle, Knight Path, Chess Attack, Go Liberties, Logic Gate Row, Domino Chain, Dice Sum, Checkers Jump, Animal Food Web, and

Compass Rose, while preserving and upgrading the seven existing puzzle types.

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## Backlog / Rejected For Now

Poker Hand Grid, Checkmate Net, Go Capture Race, Circuit Trace, Train Routes, Music Measure, Chemistry Valence, Sports Formation, Calendar Week, Language Pattern, Map Compass Walk, and Set Attribute Grid remain backlog. They need stronger domain validators or more briefing space before they are fair enough for daily play.

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## Validation Results

```
node --check public/symbols.js
node --check public/scoring.js
node --check public/puzzles.js
node --check public/game.js
node --check public/lab.js
node scripts/validate-scoring.js
  Validated lower-is-better scoring cases.
node scripts/validate-puzzles.js
  Validated 4500 daily rounds across 300 dates and 18 puzzle types.
node scripts/validate-pattern-systems.js
  Validated pattern systems: 18 types, 300 dates, 5 session attempts.
git diff --check
  Passed.
```

Browser verification covered intro, hidden board briefing, active board, wrong tap, correct feedback, transition briefings, completion, share text rendering, and lab rendering. The in-app clipboard click timed out during automation, so the report relies on the visible share text field plus existing clipboard fallback code rather than claiming a fresh automated clipboard assertion.

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## Local Run Instructions

```
python3 -m http.server 8080 -d public
# then open http://localhost:8080
# puzzle lab: http://localhost:8080/lab.html
```

The app also opens directly from `public/index.html` ; this was checked with Chrome headless DOM output.

## Netlify Deploy Settings

- Build command: leave blank
  - Publish directory: `public`
- 

## Screenshots

## Main intro screen

PUZZLE #001

# One Wrong Move

TIME  
0.0

READY

DAILY PUZZLE

MISTAKE

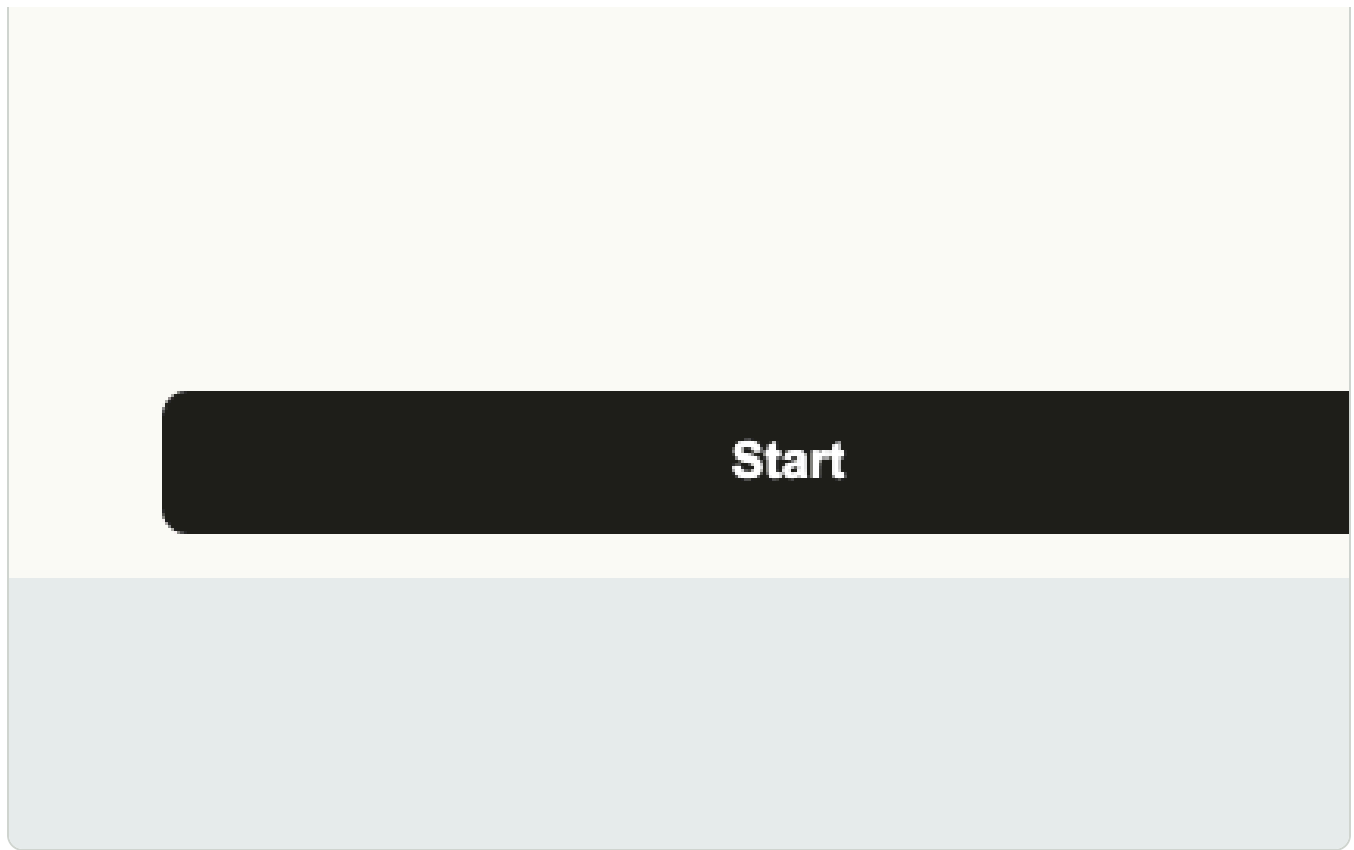
DAILY PUZZLE READY

**One board. One rule. One wrong move.**

Score = rounded-up solving time + 10s per mistake.  
Lower is better.

The board stays hidden until you open a  
round briefing.





007-main-intro-headless.png

## Round 1 briefing with new symbols

PUZZLE #001

One Wrong Move

TIME  
0.0s

ROUND 1 OF 3

VARIANT 2

MISTAKES 0

ROUND 1: DICE SUM · DICE

Rows show three dice and a total. Tap the die or total that breaks the sum.

Study the mechanic. The board stays hidden and the timer is paused.

SYMBOLS:

1

2

3

4

5

6

EXAMPLE

2

3

1

=

6

2 + 3 + 1 = 6.

Round 1 is hidden until you press Start Round.

Start Round

Restart

007-round-1-briefing-cachebusted.png

## Round 1 active board

PUZZLE #001

TIME

One Wrong Move

0.2s

ROUND 1 OF 3

VARIANT 2

MISTAKES 0

ROUND 1: DICE SUM · DICE

Rows show three dice and a total. Tap the die or total that breaks the sum.

Infer the rule from the board, then tap the one symbol that breaks it.

|   |   |   |   |    |
|---|---|---|---|----|
| 2 | 2 | 3 | = | 7  |
| 3 | 4 | 4 | = | 11 |
| 4 | 6 | 5 | = | 15 |
| 5 | 5 | 6 | = | 13 |
| 6 | 4 | 1 | = | 11 |

Restart

007-round-1-active-cachebusted.png

## Wrong-tap state



PUZZLE #001

TIME

One Wrong Move

10.1s

ROUND 1 OF 3

VARIANT 2

MISTAKES 1

ROUND 1: DICE SUM · DICE

Rows show three dice and a total. Tap the die or total that breaks the sum.

Infer the rule from the board, then tap the one symbol that breaks it.

Try again

Numbers stay small: add the first three dice and compare to the target.

|   |   |   |   |    |
|---|---|---|---|----|
| 2 | 2 | 3 | = | 7  |
| 3 | 4 | 4 | = | 11 |
| 4 | 6 | 5 | = | 15 |
| 5 | 5 | 6 | = | 13 |
| 6 | 4 | 1 | = | 11 |

Restart

007-wrong-tap-state.png

## Correct feedback

PUZZLE #001

TIME

One Wrong Move

10.2s

ROUND 1 OF 3

VARIANT 2

MISTAKES 1

ROUND 1: DICE SUM · DICE

Rows show three dice and a total. Tap the die or total that breaks the sum.

Infer the rule from the board, then tap the one symbol that breaks it.

Correct

That die or total breaks the row sum.

|   |   |   |   |    |
|---|---|---|---|----|
| 2 | 2 | 3 | = | 7  |
| 3 | 4 | 4 | = | 11 |
| 4 | 6 | 5 | = | 15 |
| 5 | 5 | 6 | = | 13 |
| 6 | 4 | 1 | = | 11 |

Next Round

Restart

007-correct-feedback.png

## Round 2 card briefing

PUZZLE #001

TIME

One Wrong Move

10.2s

ROUND 2 OF 3

VARIANT 2

MISTAKES 1

ROUND 2: SUIT CYCLE · CARDS

Suits cycle across the row and shift by row. Tap the card with the wrong suit.

Study the mechanic. The board stays hidden and the timer is paused.

SYMBOLS:

A♠

2♥

3♦

4♣

5♠

7♥

J♣

EXAMPLE

A♠

A♥

A♦

A♣

The suit cycle shifts: ♠, ♥, ♦, ♣.

Round 2 is hidden until you press Start Round.

Start Round

Restart

007-round-2-briefing.png



## Suit Cycle active board

PUZZLE #001

TIME

One Wrong Move

10.4s

ROUND 2 OF 3

VARIANT 2

MISTAKES 1

ROUND 2: SUIT CYCLE · CARDS

Suits cycle across the row and shift by row. Tap the card with the wrong suit.

Infer the rule from the board, then tap the one symbol that breaks it.

|         |         |         |         |         |
|---------|---------|---------|---------|---------|
| A<br>A♥ | A<br>A♦ | A<br>A♣ | A<br>A♠ | A<br>A♥ |
| 3<br>3♦ | 3<br>3♣ | 3<br>3♠ | 3<br>3♥ | 3<br>3♦ |
| 5<br>5♣ | 5<br>5♠ | 5<br>5♥ | 5<br>5♦ | 5<br>5♣ |
| 7<br>7♠ | 7<br>7♥ | 7<br>7♦ | 7<br>7♣ | 7<br>7♠ |
| 9<br>9♥ | 9<br>9♦ | 9<br>9♣ | A<br>A♦ | 9<br>9♥ |

Restart

007-round-2-active-card.png

## **Knight Path active board**

PUZZLE #001

TIME

One Wrong Move

10.8s

ROUND 3 OF 3

VARIANT 2

MISTAKES 1

ROUND 3: KNIGHT PATH · CHESS

Numbered knights should move in legal L-shapes. Tap the first illegal knight.

Infer the rule from the board, then tap the one symbol that breaks it.

|    |   |    |    |    |
|----|---|----|----|----|
| 1  | 6 |    | 8  | 13 |
|    |   | 2  |    | 9  |
| 5  |   | 7  | 12 |    |
| 10 |   |    | 3  |    |
|    | 4 | 11 |    |    |

Restart

007-round-3-active-knight.png

## Completion and share screen

PUZZLE #001

TIME

One Wrong Move

10.8s

COMPLETE

VARIANT 2

MISTAKES 1

DAILY PUZZLE SOLVED

Solved 3 of 3

Score = rounded-up solving time + 10s per mistake.

Lower is better.

SOLVED

3 of 3

ACTIVE TIME

10.8s

BASE TIME

11s

FINAL SCORE

21s

LOWER IS BETTER

Mistakes

1

Penalty

1 × 10s = +10s

Formula

11s + 10s = 21s

SHARE TEXT

One Wrong Move #001

Score: 21s

Lower is better

Variant 2

Base: 11s

Mistakes: 1 × 10s = +10s

✓

✓

✓

Share result

Play another mix

Restart



007-completion-share.png

## Card Straight lab board

# One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

CARDS · DIFFICULTY 1

## Card Straight

Show answer

Each row is a short straight. Tap the card that breaks rank or suit logic.

A♠2♥3♦4♣5♠7♥J♣

|         |           |           |         |           |
|---------|-----------|-----------|---------|-----------|
| 5<br>5♥ | 6<br>6♦   | 7<br>7♣   | 8<br>8♠ | 9<br>9♥   |
| 6<br>6♦ | 7<br>7♣   | 8<br>8♠   | 9<br>9♥ | 10<br>10♦ |
| 7<br>7♣ | 8<br>8♠   | 9<br>9♥   | Q<br>Q♦ | J<br>J♣   |
| 8<br>8♠ | 9<br>9♥   | 10<br>10♦ | J<br>J♣ | Q<br>Q♠   |
| 9<br>9♥ | 10<br>10♦ | J<br>J♣   | Q<br>Q♠ | K<br>K♥   |

**Break signature**

card-straight|wrong-rank|13|Q♦Q{"rank":"Q","suit":"diamond","suitIndex":1}

**Evidence**

Each row forms a five-card straight with a consistent suit cycle; the tapped card breaks rank or suit order.

**Answer**

007-lab-card-straight.png

## Suit Cycle lab board

DEVELOPMENT LAB

# One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

CARDS · DIFFICULTY 2

## Suit Cycle

Show answer

Suits cycle across the row and shift by row. Tap the card with the wrong suit.

A♠2♥3♦4♣5♠7♥J♣

|         |         |         |         |         |
|---------|---------|---------|---------|---------|
| A<br>A♥ | A<br>A♦ | A<br>A♣ | A<br>A♠ | A<br>A♥ |
| 3<br>3♦ | 3<br>3♦ | 3<br>3♣ | 3<br>3♥ | 3<br>3♦ |
| 5<br>5♣ | 5<br>5♠ | 5<br>5♥ | 5<br>5♦ | 5<br>5♣ |
| 7<br>7♣ | 7<br>7♥ | 7<br>7♦ | 7<br>7♣ | 7<br>7♣ |
| 9<br>9♥ | 9<br>9♦ | 9<br>9♣ | 9<br>9♠ | 9<br>9♥ |

**Break signature**

suit-cycle|duplicate-suit|6|3♦3{"rank":"3","suit":"diamond","suitIndex":1}

**Evidence**

The rows shift a four-suit cycle; the tapped card uses the wrong suit for its position.

**Answer**

007-lab-suit-cycle.png

## **Knight Path lab board**



DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

CHESS · DIFFICULTY 3

Show answer

Knight Path

Numbered knights should move in legal L-shapes. Tap the first illegal knight.



|    |   |    |    |    |
|----|---|----|----|----|
| 1  | 6 |    |    | 13 |
|    | 9 | 2  |    |    |
| 5  |   | 7  | 12 |    |
| 10 |   | 8  | 3  |    |
|    | 4 | 11 |    |    |

**Break signature**

knight-path|orthogonal-move|17|8

**Evidence**

The legal path establishes knight L-moves; this numbered knight is the first non-L jump.

007-lab-knight-path.png

## Chess Attack lab board

# One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

CHESS · DIFFICULTY 4

## Chess Attack

Show answer

Each piece should legally attack its target vector. Tap the illegal pair.



|  |   |      |   |    |
|--|---|------|---|----|
|  | → | 2,0  | = | OK |
|  | → | 3,-3 | = | OK |
|  | → | 2,1  | = | OK |
|  | → | 2,2  | = | OK |
|  | → | -2,0 | = | OK |

**Break signature**

chess-attack|king-illegal-target-1|17|2,2"2,2"

**Evidence**

Rows teach the legal movement families; row 4 assigns a target vector its piece cannot attack.

**Answer**

007-lab-chess-attack.png

## Go Liberties lab board

DEVELOPMENT LAB

# One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

GO · DIFFICULTY 4

## Go Liberties

Show answer

Each black stone group should show exactly two liberties. Tap the point that breaks the count.

●

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□

|              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|
| <div>○</div> | <div>+</div> | <div>●</div> | <div>+</div> | <div>○</div> |
| <div>○</div> | <div>○</div> | <div>●</div> | <div>+</div> | <div>○</div> |
| <div>○</div> | <div>+</div> | <div>●</div> | <div>+</div> | <div>○</div> |
| <div>○</div> | <div>+</div> | <div>●</div> | <div>+</div> | <div>○</div> |
| <div>○</div> | <div>+</div> | <div>●</div> | <div>+</div> | <div>○</div> |

**Break signature**

go-liberties|one-too-few-liberties|6|○"go-white"

**Evidence**

Every row shows a black stone with exactly two orthogonal liberties; the tapped point changes that count.

007-lab-go-liberties.png



## Logic Gate Row lab board

# One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

LOGIC · DIFFICULTY 2

## Logic Gate Row

Show answer

Each row is A, B, gate, equals, output. Tap the wrong output.

0

1

AND

OR

XOR

NOT

NAND

=

|   |   |      |   |   |
|---|---|------|---|---|
| 1 | 0 | AND  | = | 0 |
| 0 | 0 | OR   | = | 0 |
| 1 | 1 | XOR  | = | 1 |
| 0 | 1 | NAND | = | 1 |
| 1 | 0 | AND  | = | 0 |

**Break signature**

logic-gate-row|xor-output-wrong|14|1"1"

**Evidence**

Rows establish tiny logic circuits; row 3 has the wrong output for its gate.

**Answer**

Hidden

007-lab-logic-gate-row.png

## Domino Chain lab board

DEVELOPMENT LAB

# One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

DOMINOES · DIFFICULTY 2

## Domino Chain

Show answer

Neighboring domino halves must match. Tap the domino that breaks the chain.

123456

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 2 3 | 3 4 | 4 5 | 5 6 | 6 1 |
| 3 4 | 4 5 | 5 6 | 6 1 | 1 2 |
| 4 5 | 5 6 | 6 1 | 1 2 | 2 3 |
| 5 6 | 6 1 | 1 2 | 2 6 | 3 4 |
| 6 1 | 1 2 | 2 3 | 3 4 | 4 5 |

  
**Break signature**

domino-chain|right-side-mismatch|18|2|6{"left":2,"right":6}

**Evidence**

Neighboring domino halves match across every clean row; the tapped domino breaks that handoff.

007-lab-domino-chain.png

## Dice Sum lab board

DEVELOPMENT LAB

# One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

Regenerate

DICE · DIFFICULTY 2

## Dice Sum

Show answer

Rows show three dice and a total. Tap the die or total that breaks the sum.

123456

|   |   |   |   |    |
|---|---|---|---|----|
| 2 | 2 | 3 | = | 7  |
| 3 | 4 | 4 | = | 11 |
| 4 | 3 | 5 | = | 15 |
| 5 | 2 | 6 | = | 13 |
| 6 | 4 | 1 | = | 11 |



**Break signature**

dice-sum|die-breaks-total|11|33

**Evidence**

Each row's first three dice add to the target; the tapped cell is the only row whose math fails.

007-lab-dice-sum.png

## **Puzzle lab overview, answers hidden**

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

SESSION ATTEMPT

2026-05-02

1

Regenerate

SYMBOL GRAMMAR ·  
DIFFICULTY 1

Show answer

Rule Rows

Each row follows a shifted symbol recipe. Find the glyph that breaks its row.

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SYMBOL GRAMMAR ·  
DIFFICULTY 2

Show answer

Conveyor Shift

Each row is the previous row shifted like a conveyor belt. Find the slipped tile.

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COMPASS · DIFFICULTY 2

Show answer

Rotation Logic

Directions rotate by a consistent step. Find the one that missed the turn.

N

NE

E

SE

S

SW

W

NW

NW

NE

SE

SW

NW

N

E

S

W

N

NE

SE

SW

NW

NE

W

S

W

N

E

file:///tmp/owm-007-pattern-symbol-systems-report.html

59/67

◆

♔

◀

⦿

⦿

Break signature

rule-rows[duplicate-symbol|24|⦿"eye"

Evidence

The other rows establish a five-symbol shifted recipe; this cell uses ⦿ where the recipe expects a different symbol.

Answer

Hidden

Validator

valid

☾

⦿

◆

◀

♔

Break signature

conveyor-shift|wrong-shifted-position|12|◀"lightning"

Evidence

Every row is the previous row shifted right; the tapped cell is out of phase.

Answer

Hidden

Validator

valid

SE

SW

NW

NE

SE

Break signature

rotation-logic[opposite-direction|15|W"west"

Evidence

Rows establish a quarter-turn direction pattern; the tapped cell rotates to the wrong direction.

Answer

Hidden

Validator

valid

SYMBOL GRAMMAR · DIFFICULTY 2

Latin Trap

Show answer

Every row and column should contain one of each symbol. Find the cell causing both repeats.

☾

◆

♔

◀

⦿

|   |   |   |   |   |
|---|---|---|---|---|
| ☾ | ☀ | ◆ | ♔ | ⦿ |
| ☀ | ◆ | ♔ | ⦿ | ☾ |
| ◆ | ♔ | ⦿ | ☾ | ☀ |
| ♔ | ⦿ | ☾ | ☀ | ◆ |
| ⦿ | ☾ | ☀ | ⦿ | ♔ |

Break signature

latin-trap|missing-symbol-trap|23|⦿"eye"

Evidence

Each row and column should contain one of every symbol; the tapped cell repeats a symbol in both directions.

Answer

Hidden

Validator

valid

SYMBOL GRAMMAR · DIFFICULTY 2

Pair Pact

Show answer

Symbols travel in partner pairs. Find the symbol that breaks a pact.

☾

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⦿

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⌵

⌴

⌵

|   |   |   |   |   |
|---|---|---|---|---|
| ♔ | ◆ | • | ⌵ | ◆ |
| ☾ | ☀ | • | ⌵ | ◆ |
| ⌴ | ⌴ | • | ⦿ | ◀ |
| ♔ | ◆ | • | ⌴ | ⌴ |
| ⦿ | ◀ | • | ☾ | ☾ |

Break signature

pair-pact|unchanged-leader|24|◀"moon"

Evidence

The board repeats stable symbol pairs; the tapped symbol is not the partner established by the other rows.

Answer

Hidden

Validator

valid

MAP · DIFFICULTY 3

Path Rhythm

Show answer

A numbered path alternates movement rhythm. Find the first move that breaks the beat.

N

NE

E

SE

S

SW

W

NW

|    |    |    |     |    |
|----|----|----|-----|----|
| 1→ | 2↓ | 3→ |     |    |
|    |    | 4↓ |     |    |
|    |    | 5→ | 6↓  | 7→ |
|    |    |    |     | 8↓ |
|    |    |    | 10↓ | 9↘ |

Break signature

path-rhythm|diagonal-slip|24|9↘9

Evidence

The numbered path alternates horizontal and vertical arrows; the tapped move is the first wrong beat.

Answer

Hidden

Validator

valid

SYMBOL GRAMMAR · DIFFICULTY 3

Mirror Trap

Show answer

The right side mirrors the left, but symbols transform into paired counterparts.

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⌵

|   |   |   |   |   |
|---|---|---|---|---|
| ⌴ | ⦿ | ↔ | ◀ | ⌴ |
| ⌴ | ☾ | ↔ | ⌴ | ⌴ |
| ♔ | ☾ | ↔ | ☀ | ◆ |
| ⌵ | ⌵ | ↔ | ◆ | ◆ |
| ⦿ | ♔ | ↔ | ◆ | ◀ |

Break signature

mirror-trap|swapped-partner|8|⌴"lock"

Evidence

The left side establishes source symbols and the right side consistently transforms them; the tapped copy uses the wrong partner.

Answer

Hidden

Validator

valid

CARDS · DIFFICULTY 1

Card Straight

Show answer

Each row is a short straight. Tap the card that breaks rank or suit logic.

A♠

2♥

3♠

4♠

5♠

7♥

J♠

|    |     |     |    |     |
|----|-----|-----|----|-----|
| 5♥ | 6♠  | 7♣  | 8♠ | 9♥  |
| 6♠ | 7♣  | 8♠  | 9♥ | 10♠ |
| 7♣ | 8♠  | 9♥  | Q♠ | J♣  |
| 8♠ | 9♥  | 10♠ | J♣ | Q♠  |
| 9♥ | 10♠ | J♣  | Q♠ | K♥  |

Break signature

card-straight|wrong-rank|13|Q♠Q{"rank":"Q","suit":"diamond","suitIndex":1}

Evidence

Each row forms a five-card straight with a consistent suit cycle; the tapped card breaks rank or suit order.

Answer

Hidden

Validator

valid

CARDS · DIFFICULTY 2

Suit Cycle

Show answer

Suits cycle across the row and shift by row. Tap the card with the wrong suit.

A♠

2♥

3♠

4♠

5♠

7♥

J♠

|    |    |    |    |    |
|----|----|----|----|----|
| A♥ | A♠ | A♠ | A♠ | A♥ |
| 3♠ | 3♠ | 3♠ | 3♥ | 3♠ |
| 5♠ | 5♠ | 5♥ | 5♠ | 5♠ |
| 7♠ | 7♥ | 7♠ | 7♠ | 7♠ |
| 9♥ | 9♠ | 9♠ | 9♠ | 9♥ |

Break signature

suit-cycle|duplicate-suit|6|3♠3{"rank":"3","suit":"diamond","suitIndex":1}

Evidence

The rows shift a four-suit cycle; the tapped card uses the wrong suit for its position.

Answer

Hidden

Validator

valid

CHESS · DIFFICULTY 3

Knight Path

Show answer

Numbered knights should move in legal L-shapes.

CHESS · DIFFICULTY 4

Chess Attack

Show answer

Each piece should legally attack its target vector. Tap the illegal pair

GO · DIFFICULTY 4

Go Liberties

Show answer

Each black stone group should show exactly two

file:///tmp/owm-007-pattern-symbol-systems-report.html

60/67

liberties. Tap the point that breaks the count.

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|  |  |  |  |  |

**Break signature**  
go-liberties|one-too-few-liberties|6|o|"go-white"

**Evidence**  
Every row shows a black stone with exactly two orthogonal liberties; the tapped point changes that count.

**Answer**  
Hidden

**Validator**  
valid

DICE · DIFFICULTY 2

Dice Sum

Show answer

Rows of three dice and a total. Tap the die or total that breaks the sum.

1

2

3

4

5

6

4

2

3

=

9

5

4

4

=

13

6

3

5

=

17

1

2

6

=

9

2

4

1

=

7

Break signature

dice-sum[die-breaks-total]11|33

Evidence

Each row's first three dice add to the target; the tapped cell is the only row whose math fails.

Answer

Hidden

Validator

valid

**COMPASS > DIFFICULTY 2**

**Compass Rose**

**Show answer**

Directions rotate by the same amount in each row.  
Tap the direction that misses the rotation.

|    |    |    |    |    |    |   |    |
|----|----|----|----|----|----|---|----|
| N  | NE | E  | SE | S  | SW | W | NW |
| E  | S  | W  | N  | E  |    |   |    |
| SE | SW | NW | NE | NW |    |   |    |
| S  | W  | N  | E  | S  |    |   |    |
| SW | NW | NE | SE | SW |    |   |    |
| W  | N  | E  | S  | W  |    |   |    |

**Break algorithm**

break signature

checkers-jump|wrong-diagonal-direction|8|3

Evidence

The numbered checkers establish diagonal movement; the tapped checker is the first off-diagonal step.

Answer

Hidden

Validator

valid

Break signature

animal-food-web|wrong-food-chain-position|18|RB"rabbit"

Evidence

The rows repeat an obvious food chain; the tapped animal is in the wrong habitat or chain position.

Answer

Hidden

Validator

valid

break signature

compass-rose|opposite-direction|9|NW"northwest"

Evidence

The compass rose turns by a fixed amount; this marker uses the wrong bearing.

Answer

Hidden

Validator

valid

007-lab-overview-hidden.png

## **Puzzle lab overview, answers shown**

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

SESSION ATTEMPT

2026-05-02

1

Regenerate

SYMBOL GRAMMAR ·  
DIFFICULTY 1

Hide answer

Rule Rows

Each row follows a shifted symbol recipe. Find the glyph that breaks its row.

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SYMBOL GRAMMAR ·  
DIFFICULTY 2

Hide answer

Conveyor Shift

Each row is the previous row shifted like a conveyor belt. Find the slipped tile.

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COMPASS · DIFFICULTY 2

Hide answer

Rotation Logic

Directions rotate by a consistent step. Find the one that missed the turn.

N

NE

E

SE

S

SW

W

NW

NW

NE

SE

SW

NW

N

E

S

W

N

NE

SE

SW

NW

NE

W

S

W

N

E



◆

♔

◀

⦿

⦿

**Break signature**  
rule-rows[duplicate-symbol|24|⦿"eye"]

**Evidence**  
The other rows establish a five-symbol shifted recipe; this cell uses ⦿ where the recipe expects a different symbol.

**Answer**  
Cell 25: That glyph breaks the shifted row grammar.

**Validator**  
valid

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⦿

◆

◀

♔

**Break signature**  
conveyor-shift|wrong-shifted-position|12|◀"lightning"

**Evidence**  
Every row is the previous row shifted right; the tapped cell is out of phase.

**Answer**  
Cell 13: That tile does not follow the conveyor shift.

**Validator**  
valid

SE

SW

NW

NE

SE

**Break signature**  
rotation-logic[opposite-direction|15|W"west"]

**Evidence**  
Rows establish a quarter-turn direction pattern; the tapped cell rotates to the wrong direction.

**Answer**  
Cell 16: That direction misses the rotation sequence.

**Validator**  
valid

SYMBOL GRAMMAR ·  
DIFFICULTY 2

Hide answer

**Latin Trap**

Every row and column should contain one of each symbol. Find the cell causing both repeats.

☾

◆

♔

◀

⦿

|   |   |   |   |   |
|---|---|---|---|---|
| ☾ | ☀ | ◆ | ♔ | ⦿ |
| ☀ | ◆ | ♔ | ⦿ | ☾ |
| ◆ | ♔ | ⦿ | ☾ | ☀ |
| ♔ | ⦿ | ☾ | ☀ | ◆ |
| ⦿ | ☾ | ☀ | ◆ | ♔ |

**Break signature**  
latin-trap|missing-symbol-trap|23|⦿"eye"

**Evidence**  
Each row and column should contain one of every symbol; the tapped cell repeats a symbol in both directions.

**Answer**  
Cell 24: That cell creates the row and column duplicate.

**Validator**  
valid

SYMBOL GRAMMAR ·  
DIFFICULTY 2

Hide answer

**Pair Pact**

Symbols travel in partner pairs. Find the symbol that breaks a pact.

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☐

|   |   |   |   |   |
|---|---|---|---|---|
| ♔ | ◆ | • | ♚ | ◆ |
| ☾ | ☀ | • | ♚ | ◆ |
| ☐ | ☐ | • | ⦿ | ◀ |
| ♔ | ◆ | • | ☐ | ☐ |
| ⦿ | ◀ | • | ☾ | ☾ |

**Break signature**  
pair-pact|unchanged-leader|24|◀"moon"

**Evidence**  
The board repeats stable symbol pairs; the tapped symbol is not the partner established by the other rows.

**Answer**  
Cell 25: That symbol breaks the pairing pact.

**Validator**  
valid

MAP · DIFFICULTY 3

Hide answer

**Path Rhythm**

A numbered path alternates movement rhythm. Find the first move that breaks the beat.

N

NE

E

SE

S

SW

W

NW

|    |    |    |     |    |
|----|----|----|-----|----|
| 1→ | 2↓ | 3→ |     |    |
|    |    | 4↓ |     |    |
|    |    | 5→ | 6↓  | 7→ |
|    |    |    |     | 8↓ |
|    |    |    | 10↓ | 9↘ |

**Break signature**  
path-rhythm|diagonal-slip|24|9↘9

**Evidence**  
The numbered path alternates horizontal and vertical arrows; the tapped move is the first wrong beat.

**Answer**  
Cell 25: That move breaks the path rhythm.

**Validator**  
valid

SYMBOL GRAMMAR ·  
DIFFICULTY 3

Hide answer

**Mirror Trap**

The right side mirrors the left, but symbols transform into paired counterparts.

☾

◆

♔

◀

⦿

☀

♚

☐

☐

|   |   |   |   |   |
|---|---|---|---|---|
| ☐ | ⦿ | ↔ | ◀ | ☐ |
| ☐ | ☾ | ↔ | ☐ | ☐ |
| ♔ | ☾ | ↔ | ☀ | ◆ |
| ♚ | ♚ | ↔ | ◆ | ◆ |
| ⦿ | ♔ | ↔ | ◆ | ◀ |

**Break signature**  
mirror-trap|swapped-partner|8|☐"lock"

**Evidence**  
The left side establishes source symbols and the right side consistently transforms them; the tapped copy uses the wrong partner.

**Answer**  
Cell 9: That mirrored symbol uses the wrong counterpart.

**Validator**  
valid

CARDS · DIFFICULTY 1

Hide answer

**Card Straight**

Each row is a short straight. Tap the card that breaks rank or suit logic.

A♠

2♥

3♠

4♠

5♠

7♥

J♠

|    |     |     |    |     |
|----|-----|-----|----|-----|
| 5♥ | 6♠  | 7♣  | 8♠ | 9♥  |
| 6♠ | 7♣  | 8♠  | 9♥ | 10♠ |
| 7♣ | 8♠  | 9♥  | Q♠ | J♣  |
| 8♠ | 9♥  | 10♠ | J♣ | Q♠  |
| 9♥ | 10♠ | J♣  | Q♠ | K♥  |

**Break signature**  
card-straight|wrong-rank|13|Q♠Q"rank":"Q","suit":"diamond","suitIndex":1}

**Evidence**  
Each row forms a five-card straight with a consistent suit order; the tapped card breaks rank or suit order.

**Answer**  
Cell 14: That card breaks the straight.

**Validator**  
valid

CARDS · DIFFICULTY 2

Hide answer

**Suit Cycle**

Suits cycle across the row and shift by row. Tap the card with the wrong suit.

A♠

2♥

3♠

4♠

5♠

7♥

J♠

|    |    |    |    |    |
|----|----|----|----|----|
| A♥ | A♠ | A♣ | A♠ | A♥ |
| 3♠ | 3♠ | 3♠ | 3♥ | 3♠ |
| 5♠ | 5♠ | 5♥ | 5♠ | 5♠ |
| 7♠ | 7♥ | 7♠ | 7♣ | 7♠ |
| 9♥ | 9♠ | 9♠ | 9♠ | 9♥ |

**Break signature**  
suit-cycle|duplicate-suit|6|3♠3"rank":"3","suit":"diamond","suitIndex":1}

**Evidence**  
The rows shift a four-suit cycle; the tapped card uses the wrong suit for its position.

**Answer**  
Cell 7: That card is off the suit cycle.

**Validator**  
valid

CHESS · DIFFICULTY 3

Hide answer

**Knight Path**

Numbered knights should move in legal L-shapes.

CHESS · DIFFICULTY 4

Hide answer

**Chess Attack**

Each piece should legally attack its target vector. Tap the illegal pair

GO · DIFFICULTY 4

Hide answer

**Go Liberties**

Each black stone group should show exactly two liberties. Tap the point that breaks the count

Tap the first illegal knight.

|    |   |    |    |    |
|----|---|----|----|----|
| 1  | 6 |    |    | 13 |
|    |   | 9  | 2  |    |
| 5  |   | 7  | 12 |    |
| 10 |   | 8  | 3  |    |
|    | 4 | 11 |    |    |

**Break signature**  
knight-path[orthogonal-move]17|8

**Evidence**  
The legal path establishes knight L-moves; this numbered knight is the first non-L jump.

**Answer**  
Cell 18: That knight jumps illegally.

**Validator**  
valid

Tap the first illegal knight.

|   |      |   |    |
|---|------|---|----|
| → | -1,0 | = | OK |
| → | 2,2  | = | OK |
| → | 1,-2 | = | OK |
| → | 2,2  | = | OK |
| → | 0,4  | = | OK |

**Break signature**  
chess-attack[king-illegal-target-1]17|2,2,"2,2"

**Evidence**  
Rows teach the legal movement families; row 4 assigns a target vector its piece cannot attack.

**Answer**  
Cell 18: That piece cannot attack the shown target.

**Validator**  
valid

Tap the first illegal knight.

|   |   |   |  |
|---|---|---|--|
| + | • | + |  |
| + | • | + |  |
| + | • | + |  |
| + | • | + |  |
| + | • | + |  |

**Break signature**  
go-liberties[one-too-few-liberties]6|○"go-white"

**Evidence**  
Every row shows a black stone with exactly two orthogonal liberties; the tapped point changes that count.

**Answer**  
Cell 7: That point gives the black stone group the wrong liberty count.

**Validator**  
valid

LOGIC · DIFFICULTY 2  
**Logic Gate Row** Hide answer

Each row is A, B, gate, equals, output. Tap the wrong output.

|   |   |      |   |   |
|---|---|------|---|---|
| 0 | 1 | AND  | = | 0 |
| 0 | 0 | OR   | = | 0 |
| 1 | 1 | XOR  | = | 1 |
| 0 | 1 | NAND | = | 1 |
| 1 | 0 | AND  | = | 0 |

**Break signature**  
logic-gate-row[xor-output-wrong]14|1"1"

**Evidence**  
Rows establish tiny logic circuits; row 3 has the wrong output for its gate.

**Answer**  
Cell 15: That output is wrong for the gate.

**Validator**  
valid

DOMINOES · DIFFICULTY 2  
**Domino Chain** Hide answer

Neighboring domino halves must match. Tap the domino that breaks the chain.

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 3 4 | 4 5 | 5 6 | 6 1 | 1 2 |
| 4 5 | 5 6 | 6 1 | 1 2 | 2 3 |
| 5 6 | 6 1 | 1 2 | 2 3 | 3 4 |
| 6 1 | 1 2 | 2 3 | 3 1 | 4 5 |
| 1 2 | 2 3 | 3 4 | 4 5 | 5 6 |

**Break signature**  
domino-chain[right-side-mismatch]18|3|1["left":3,"right":1]

**Evidence**  
Neighboring domino halves match across every clean row; the tapped domino breaks that handoff.

**Answer**  
Cell 19: That domino breaks the matching chain.

**Validator**  
valid

DICE · DIFFICULTY 2  
**Dice Sum** Hide answer

Rows show three dice and a total. Tap the die or total that breaks the sum.

|   |   |   |   |    |
|---|---|---|---|----|
| 4 | 2 | 3 | = | 9  |
| 5 | 4 | 4 | = | 13 |
| 6 | 3 | 5 | = | 17 |
| 1 | 2 | 6 | = | 9  |
| 2 | 4 | 1 | = | 7  |

**Break signature**  
dice-sum[die-breaks-total]11|33

**Evidence**  
Each row's first three dice add to the target; the tapped cell is the only row whose math fails.

**Answer**  
Cell 12: That die or total breaks the row sum.

**Validator**  
valid

CHECKERS · DIFFICULTY 3  
**Checkers Jump** Hide answer

Numbered checkers must step diagonally. Tap the first illegal move.

|   |   |   |   |   |
|---|---|---|---|---|
|   | 1 |   |   |   |
| 8 |   | 2 | 3 |   |
|   | 7 |   |   |   |
|   |   | 6 |   | 4 |
|   |   |   | 5 |   |

**Break signature**  
checkers-jump[diagonal-move]17|8

**Evidence**  
Rows show numbered checkers; row 2 shows a diagonal move that is illegal.

**Answer**  
Cell 18: That move is illegal.

**Validator**  
valid

ECOLOGY · DIFFICULTY 2  
**Animal Food Web** Hide answer

Each row shows an obvious food chain. Tap the animal that does not belong.

|    |    |    |    |    |
|----|----|----|----|----|
| PL | IN | FR | SN | HW |
| PL | IN | FR | SN | HW |
| PL | IN | FR | SN | HW |
| PL | IN | FR | RB | HW |
| PL | IN | FR | SN | HW |

**Break signature**  
ecology-food-web[animal-not-in-chain]18|3|1["left":3,"right":1]

**Evidence**  
Rows show a food chain; row 4 shows a predator that does not belong.

**Answer**  
Cell 19: That animal breaks the matching chain.

**Validator**  
valid

COMPASS · DIFFICULTY 2  
**Compass Rose** Hide answer

Directions rotate by the same amount in each row. Tap the direction that misses the rotation.

|    |    |    |    |    |
|----|----|----|----|----|
| E  | S  | W  | N  | E  |
| SE | SW | NW | NE | NW |
| S  | W  | N  | E  | S  |
| SW | NW | NE | SE | SW |
| W  | N  | E  | S  | W  |

**Break signature**  
compass-rose[direction-not-in-rotation]18|3|1["left":3,"right":1]

**Evidence**  
Rows show a compass rose; row 2 shows a direction that does not belong.

**Answer**  
Cell 19: That direction breaks the matching chain.

**Validator**  
valid

|                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>break signature</div><div>checkers-jump wrong-diagonal-direction 8 3</div></div> <div><div>Evidence</div><div>The numbered checkers establish diagonal movement; the tapped checker is the first off-diagonal step.</div></div> <div><div>Answer</div><div>Cell 9: That checker does not move diagonally.</div></div> <div><div>Validator</div><div>valid</div></div> | <div><div>Break signature</div><div>animal-food-web wrong-food-chain-position 18 RB"rabbit"</div></div> <div><div>Evidence</div><div>The rows repeat an obvious food chain; the tapped animal is in the wrong habitat or chain position.</div></div> <div><div>Answer</div><div>Cell 19: That animal breaks the food chain.</div></div> <div><div>Validator</div><div>valid</div></div> | <div><div>break signature</div><div>compass-rose opposite-direction 9 NW"northwest"</div></div> <div><div>Evidence</div><div>The compass rose turns by a fixed amount; this marker uses the wrong bearing.</div></div> <div><div>Answer</div><div>Cell 10: That compass marker misses the rotation.</div></div> <div><div>Validator</div><div>valid</div></div> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

007-lab-overview-answer-shown.png